Calculus II

Solution to Assignment 2

Part I. Regular problems

1. (a) Let $f(x) = \frac{2}{x \ln x}$. Then f is continuous and positive on $[2, \infty)$ and

$$f'(x) = \frac{-2(1 + \ln x)}{x^2 (\ln x)^2} < 0$$

for $x \in [2, \infty)$, so f is decreasing. Thus we can use the Integral Test,

$$\int_2^\infty \frac{2}{x \ln x}\, dx = \lim_{R \to \infty} \int_2^R \frac{2}{x \ln x}\, dx = 2 \lim_{R \to \infty} [\ln(\ln x)]_2^R = 2 \lim_{R \to \infty} [\ln(\ln R) - \ln(\ln 2)] = \infty,$$

so $\sum_{k=2}^\infty \frac{2}{k \ln k}$ diverges.

(b)

$$\ln(k!) = \ln(1 \cdot 2 \cdot 3 \cdot \ldots \cdot (k-1) \cdot k) = \ln 1 + \ln 2 + \ln 3 + \ldots + \ln(k-1) + \ln k$$

Because $\ln(k!) \le k \ln k$, $\frac{1}{k \ln k} \le \frac{1}{\ln(k!)}$. In (a), it is proved that the series $\sum_{k=2}^\infty \frac{1}{k \ln k}$ diverges. Therefore, by Limit Comparison Test, $\sum_{k=2}^\infty \frac{1}{\ln(k!)}$ also diverges.

(c) First note that

$$\frac{k e^{-k^2}}{4 + e^{-k}} = \frac{k}{4e^{k^2} + e^{k^2 - k}} \le \frac{k}{4e^{k^2}}.$$

If we show convergence of $\sum_{k=1}^\infty \frac{k}{4e^{k^2}}$, then convergence of $\sum_{k=1}^\infty \frac{k e^{-k^2}}{4 + e^{-k}}$ will follow by the Comparison Test. Let $f(x) = \frac{x}{4e^{x^2}}$. Since $f'(x) = \frac{1 - 2x^2}{4e^{x^2}} \le 0$ for $x \ge 1$, we can use the Integral Test.

$$\int_1^\infty \frac{x\, dx}{4e^{x^2}} = \lim_{R \to \infty} \int_1^R \frac{x\, dx}{4e^{x^2}} = \lim_{R \to \infty} \left[-\frac{1}{8e^{x^2}} \right]_1^R = \lim_{R \to \infty} \left(-\frac{1}{8e^{R^2}} + \frac{1}{8e} \right) = \frac{1}{8e}.$$

Thus $\sum_{k=1}^\infty \frac{k}{4e^{k^2}}$ converges.

(d) Since $0 \le \frac{\sin^{-1}(1/k)}{k^2} \le \frac{\pi}{2k^2}$ for all $k \ge 1$ and $\sum_{k=1}^\infty \frac{\pi}{2k^2}$ is a convergent p-series, $\sum_{k=1}^\infty \frac{\sin^{-1}(1/k)}{k^2}$ is convergent, by the Comparison Test.

(e) Since $\left| \frac{\cos(k\pi/s)}{k!} \right| \le \frac{1}{k!}$ and $\sum_{k=1}^\infty \frac{1}{k!}$ is convergent by the Ratio Test. Therefore, by the Comparison Test, $\sum_{k=1}^\infty \frac{\cos(k\pi/s)}{k!}$ is absolutely convergent and hence convergent.

2. (a) Since $\sum_{k=1}^{\infty} \left|\frac{(-1)^{k+1}}{k^4}\right| = \sum_{k=1}^{\infty} \frac{1}{k^4}$ and $\sum_{k=1}^{\infty} \frac{1}{k^4}$ converges, $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^4}$ converges absolutely and hence converges.

   (b) We then approximate the sum by

$$S \approx S_{40} \approx 0.9470326439.$$

   From error estimate in Alternating Seires Test, we have

$$|S - S_{40}| \leq a_{41} = \frac{1}{41^4} \approx 3.54 \times 10^{-7}.$$

   This says that our approximation $S \approx 0.9470326439$ is off by no more than $\pm 3.54 \times 10^{-7}$.

   (c) In this case, we want to find the number of terms n for which

$$|S - S_n| \leq 1 \times 10^{-10}.$$

   We have that

$$|S - S_n| \leq a_{n+1} = \frac{1}{(n+1)^4}.$$

   So, we look for n such that

$$\frac{1}{(n+1)^4} \leq 1 \times 10^{-10}.$$

   Solving for n, we get

$$10^{10} \leq (n+1)^4,$$

   so that

$$\sqrt[4]{10^{10}} \leq n+1$$

   or

$$n \geq \sqrt[4]{10^{10}} - 1 \approx 315.2.$$

   So, if we take $n \geq 316$, we will guarantee an error of no more than 1×10^{-10}. Using the suggested number of terms, we get the approximate sum

$$S \approx S_{316} \approx 0.947032829447,$$

   which we now know to be correct to within 1×10^{-10}.

3. Suppose that $\sum_{k=0}^{\infty} a_k$ converges, then $a_k > 0$ gives $\frac{a_k}{1+a_k} \leq a_k$, so that with Limit Comparison Test, $\sum_{k=0}^{\infty} \frac{a_k}{1+a_k}$ converges.

   On the contrary, suppose that $\sum_{k=0}^{\infty} \frac{a_k}{1+a_k}$ converges. Let $b_k = \frac{a_k}{1+a_k}$, then $a_k = \frac{b_k}{1-b_k}$. Then we have $\lim_{k\to\infty} b_k = 0$, so that for a big enough positive integer N, we have $b_k < \frac{1}{2}$ for $k \geq N$. Then for $k \geq N$, $1 - b_k > \frac{1}{2}$, and then $a_k < 2b_k$. Thus using the Limit Comparison Test, the series $\sum_{k=N}^{\infty} a_k$ converges, and so does $\sum_{k=0}^{\infty} a_k$.

4. (a) Since $\lim_{k\to\infty} \sqrt[k]{|a_k|} = \lim_{k\to\infty} \sqrt[k]{\left|\left(\frac{1-3k}{4k}\right)^k\right|} = \lim_{k\to\infty} \left|\frac{1-3k}{4k}\right| = \frac{3}{4} < 1$, $\sum_{k=5}^{\infty} \left(\frac{1-3k}{4k}\right)^k$ converges, by the Root Test.

   (b) Since $\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \frac{(k+1)^3 e^{-(k+1)}}{k^3 e^{-k}} = \lim_{k\to\infty} \left(\frac{k+1}{k}\right)^3 \frac{1}{e} = \frac{1}{e} < 1$, $\sum_{k=1}^{\infty} k^3 e^{-k}$ converges absolutely, by the Ratio Test.

   (c) $\lim_{k\to\infty} a_k = \lim_{k\to\infty} \left(\frac{e^k}{k^2}\right)^k = \infty$, so $\sum_{k=4}^{\infty} \left(\frac{e^k}{k^2}\right)^k$ diverges by the k-th Term Test for Divergence.

   (d) Using the Ratio Test,

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \frac{(k+1)^{10} \cdot 2k!}{(2k+2)! \cdot k^{10}} = \lim_{k\to\infty} \frac{(k+1)^{10}}{k^{10}(2k+1)(2k+2)}$$
$$= \lim_{k\to\infty} \frac{(k+1)^{10}}{4k^2 + 6k^{11} + 2k^{10}} = 0 < 1$$

   so $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^{10}}{(2k)!}$ converges absolutely.

   (e) Since

$$\lim_{k\to\infty} \frac{a_{k+1}}{a_k} = \lim_{k\to\infty} \frac{(k+1)! \cdot 1 \cdot 3 \cdot 5 \cdots (2k-1)}{1 \cdot 3 \cdot 5 \cdots (2k+1) \cdot k!} = \lim_{k\to\infty} \frac{k+1}{2k+1} < 1,$$

   proving convergence of the series $\sum_{k=1}^{\infty} \frac{k!}{1 \cdot 3 \cdot 5 \cdots (2k-1)}$, by the Ratio Test.

   (f) Since

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \frac{(k+1)^2 4^{k+1}}{(k+1)!} \cdot \frac{k!}{k^2 4^k} = \lim_{k\to\infty} \frac{4(k+1)}{k^2} = 0 < 1,$$

   $\sum_{k=4}^{\infty} (-1)^{k+1} \frac{k^2 4^k}{k!}$ converges absolutely, by the Ratio Test.

   (g) $\lim_{k\to\infty} a_k = \lim_{k\to\infty} \frac{k}{2k+1} = \frac{1}{2}$, so by the k-th Term Test for Divergence, $\sum_{k=2}^{\infty} (-1)^{k+1} \frac{k}{2k+1}$ diverges.

Part II. Group problems

1. (a) $-\frac{2}{5} + \frac{4}{6} - \frac{6}{7} + \frac{8}{8} - \frac{10}{9} + \cdots = \sum_{n=1}^{\infty} (-1)^n \frac{2n}{n+4}$. Now $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{2n}{n+4} = \lim_{n\to\infty} \frac{2}{1+4/n} = \frac{2}{1} \neq 0$. Since $\lim_{n\to\infty} a_n \neq 0$ (in fact the limit does not exist), the series diverges by the Test for Divergence.

   (b) $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n+1}} = \sum_{n=0}^{\infty} (-1)^{n+1} b_n$. Now $b_n = \frac{1}{\sqrt{n+1}} > 0$, $\{b_n\}$ is decreasing, and $\lim_{n\to\infty} b_n = 0$, so the series converges by the Alternating Series Test.

(c) $b_n = \frac{n}{2^n} > 0$ for $n \geq 1$. $\{b_n\}$ is decreasing for $n \geq 2$ since $\left(\frac{x}{2^x}\right)' = \frac{2^x - x2^x \ln 2}{(2^x)^2} = \frac{1 - x\ln 2}{2^x} < 0$ for $x > \frac{1}{\ln 2} \approx 1.4$. Also, $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n}{2^n} \overset{H}{=} \lim_{n\to\infty} \frac{1}{2^n \ln 2} = 0$. Thus, the series $\sum_{n=1}^{\infty} (-1)^n \frac{n}{2^n}$ converges by the Alternating Series Test.

2. (a) Using the Ration Test,

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k\to\infty} \frac{(k+1)2^{k+1}}{3^{k+1}} \cdot \frac{3^k}{k2^k} = \lim_{k\to\infty} \frac{2(k+1)}{3k} = \frac{2}{3} < 1,$$

so $\sum_{k=6}^{\infty} (-1)^k \frac{k2^k}{3^k}$ converges absolutely.

(b) By the Alternating Series Test,

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} \frac{k}{k^2+1} = 0$$

and

$$\frac{a_{k+1}}{a_k} = \frac{k+1}{(k+1)^2+1} \cdot \frac{k^2+1}{k} = \frac{k^3+k^2+k+1}{k^3+2k^2+2k} < 1$$

for all $k \geq 1$, so $a_{k+1} < a_k$ for all $k \geq 1$, so the series converges. But by the Limit Comparison Test, letting $a_k = \frac{k}{k^2+1}$ and $b_k = \frac{1}{k}$,

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \frac{k^2}{k^2+1} = 1 > 0$$

and $\sum_{k=1}^{\infty} \frac{1}{k}$ is the divergent harmonic series. Therefore $\sum_{k=1}^{\infty} |a_k| = \sum_{k=1}^{\infty} \frac{k}{k^2+1}$ diverges. Thus $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k}{k^2+1}$ converges conditionally.

(c) Since, for $k \geq 2$, $a_k = \frac{1}{k\ln k}$ is a decreasing sequence of positive terms and

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} \frac{1}{k\ln k} = 0,$$

the series $\sum_{k=2}^{\infty} \frac{(-1)^k}{k\ln k}$ is convergent, by the Alternating Series Test.
On the other hand, since the function $f(x) = \frac{1}{x\ln x}$ is positive, decreasing on $[2,\infty)$, and

$$\int_2^{\infty} f(x)\,dx = \int_2^{\infty} \frac{1}{x\ln x}\,dx = \infty,$$

the series $\sum_{k=2}^{\infty} \frac{1}{k\ln k}$ is divergent, by the Integral Test. Thus, $\sum_{k=2}^{\infty} \frac{(-1)^k}{k\ln k}$ is conditionally convergent.